DATA SCIENCE PROJECT REPORT

Supermarket Sales EDA

Univariate · Bivariate · Multivariate Analysis

Branch Performance, Customer Behavior & Sales Patterns

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Executive Summary

This report documents a comprehensive Exploratory Data Analysis (EDA) on a supermarket sales dataset from Myanmar, spanning January–March 2019. The project systematically applies three levels of analysis — univariate, bivariate, and multivariate — to extract business insights on branch performance, customer demographics, product lines, payment preferences, and temporal sales patterns.

The analysis showcases advanced Pandas and Seaborn techniques including cross-tabulation, datetime feature engineering, heatmaps, grouped visualizations with hue, and distribution skewness measurement — skills directly relevant to business intelligence and data analytics roles.

Project Highlights

Metric	Value
Dataset Size	1,000 rows × 17 columns
Data Quality	Zero null values across all columns
Time Period	January 2019 – March 2019
Branches	3 (A-Yangon, B-Mandalay, C-Naypyitaw)

Metric	Value
Product Lines	6 categories
Payment Methods	3 (Ewallet, Cash, Credit Card)
Average Unit Price	$55.67
Average Rating (1–10 scale)	Normally distributed, skewness ≈ 0.009
Most Profitable Branch	Branch C (Naypyitaw)
Highest-Income Product Line	Home and Lifestyle
Peak Sales Day	Saturday
Peak Sales Month	January
Analysis Types	Univariate (4), Bivariate/Multivariate (10)
Tools Used	Python, Pandas, NumPy, Seaborn, Matplotlib

1. Introduction

1.1 Project Background

Understanding retail sales data is a foundational skill in data analytics. This project uses a publicly available supermarket transaction dataset from three Myanmar branches to practice the full EDA workflow — from data profiling through multi-dimensional analysis using Python.

The unique value of this project over basic EDA is its explicit application of all three analysis levels: univariate (one variable at a time), bivariate (two variables), and multivariate (three or more variables simultaneously), making it a comprehensive portfolio demonstration.

1.2 Objectives

- Profile the dataset structure, types, and quality
- Separate columns into categorical and numerical programmatically
- Perform univariate analysis on key columns (Branch, Payment, Rating, cogs)
- Perform bivariate analysis across numerical–numerical and numerical–categorical pairs
- Perform multivariate analysis using hue-based visualizations and cross-tabulations
- Engineer new features from datetime columns (day of week, month)
- Answer 14 business questions from the dataset

1.3 The 14 Analysis Questions

#	Question	Analysis Type	Chart Used
1	Distribution of sales by branch	Univariate – Categorical	countplot, bar chart

#	Question	Analysis Type	Chart Used
2	Most popular payment method	Univariate – Categorical	bar chart
3	Distribution of customer ratings	Univariate – Numerical	distplot (KDE)
4	Distribution of cost of goods sold (cogs)	Univariate – Numerical	distplot, boxplot
5	Does cogs affect customer ratings?	Bivariate – Num–Num	scatterplot
6	Does gross income affect ratings?	Bivariate – Num–Num	scatterplot
7	Most profitable branch by gross income	Bivariate – Num–Cat	barplot
8	Gender vs. gross income relationship	Bivariate – Num–Cat	boxplot
9	Product line with most income	Bivariate – Num–Cat	barplot (rotated x)
10	Highest unit price by product line	Bivariate – Num–Cat	barplot (rotated x)
11	Payment method usage by city	Multivariate – Cat–Cat	crosstab + heatmap
12	Product line with highest quantity sold	Multivariate – Num–Cat	groupby + bar chart
13	Daily sales by day of week	Time Series	bar chart (engineered col)
14	Monthly sales trend	Time Series	bar chart (engineered col)

2. Dataset Overview

2.1 Data Source

The dataset is the Supermarket Sales dataset from Kaggle, representing real transaction records from a supermarket chain with branches in Yangon, Mandalay, and Naypyitaw, Myanmar. Data covers January 1 to March 30, 2019.

Repository: github.com/princemaheta2627-glitch/univariate_bivariate_multivariate_analysis_eda

2.2 Dataset Structure

The dataset contains 1,000 rows and 17 columns, all fully populated with zero null values — a clean dataset ready for analysis without preprocessing overhead.

Column	Type	Unique Values	Description
Invoice ID	object	1,000	Unique transaction ID
Branch	object (cat)	3	Branch: A (Yangon), B (Mandalay), C (Naypyitaw)
City	object (cat)	3	City of the branch
Customer type	object (cat)	2	Member (loyalty card) or Normal
Gender	object (cat)	2	Male or Female
Product line	object (cat)	6	Category of product purchased
Unit price	float64 (num)	943	Price per product unit in USD
Quantity	int64 (cat)	10	Number of items purchased (1–10)
Tax 5%	float64 (num)	990	5% tax applied on purchase
Total	float64 (num)	990	Total price including tax
Date	datetime64 (num)	89	Purchase date (Jan–Mar 2019)
Time	object (num)	506	Purchase time (10:00–21:00)
Payment	object (cat)	3	Cash, Credit card, or Ewallet
cogs	float64 (num)	990	Cost of Goods Sold
gross margin percentage	float64 (cat)	1	Constant 4.76% for all rows
gross income	float64 (num)	990	Gross profit per transaction
Rating	float64 (num)	61	Customer experience rating (1–10)

2.3 Categorical vs. Numerical Classification

A key skill demonstrated in this project is automated column classification using a threshold-based approach:

```
– for column in data1.columns:
–     if data1[column].nunique() > 10:
–         num.append(column)
–     else:
–         cat.append(column)
```

Class	Columns
Categorical (≤10 unique)	Branch, City, Customer type, Gender, Product line, Quantity, Payment, gross margin percentage

Class	Columns
Numerical (>10 unique)	Invoice ID, Unit price, Tax 5%, Total, Date, Time, cogs, gross income, Rating

Note: This programmatic classification is practically useful — it handles any dataset size without manual inspection, though domain knowledge is still applied to interpret the results correctly.

3. Methodology

3.1 Tools & Libraries

Library	Role	Key Usage
Python 3	Core language	All logic, loops, custom mappings
Pandas	Data manipulation	groupby, crosstab, value_counts, dt accessor, map()
NumPy	Numerical support	Background array operations
Seaborn	Statistical visualization	countplot, barplot, boxplot, scatterplot, heatmap, distplot
Matplotlib	Plot rendering	xticks(rotation=60), show(), axis labels
Kaggle Notebook	Environment	Jupyter-based cloud execution

3.2 Analysis Framework

The project follows a structured three-level EDA framework:

Level	Definition	Variables Involved	Techniques Used
Univariate	Analysis of one variable at a time	1 column	countplot, bar chart, distplot, boxplot, skew()
Bivariate	Relationship between two variables	2 columns	scatterplot, barplot, boxplot with x/y
Multivariate	Interactions among 3+ variables	3+ columns	hue parameter, crosstab, heatmap, groupby aggregation

3.3 Feature Engineering

Two new features were created from the Date column using Pandas datetime accessors:

- `data1['day_of_week'] = data1['Date'].dt.dayofweek.map({0:'Mon',1:'Tue',2:'Wed',3:'Thurs',4:'Fri',5:'Sat',6:'Sun'})`
- `data1['month'] = data1['Date'].dt.month.map({1:'Jan',2:'Feb',3:'Mar',...})`

These engineered features enabled temporal pattern analysis not available from the raw date string.

4. Univariate Analysis — Findings

4.1 Branch Sales Distribution (Categorical)

A countplot and bar chart of the Branch column revealed near-equal transaction counts across all three branches:

Branch	City	Transaction Count	Share
A	Yangon	340	34.0%
B	Mandalay	332	33.2%
C	Naypyitaw	328	32.8%

Insight: Remarkably balanced traffic across all branches — no single branch dominates in transaction volume. This suggests uniform marketing and operational capacity.

4.2 Payment Method Distribution (Categorical)

Payment Method	Count	Share
Ewallet	345	34.5%
Cash	344	34.4%
Credit Card	311	31.1%

Insight: Ewallet is the most popular payment method — very narrowly ahead of Cash. Credit Card lags slightly, suggesting an opportunity for targeted promotions to increase card adoption.

4.3 Rating Distribution (Numerical)

The customer rating column (scale 1–10) was analyzed using a distplot with KDE overlay:

- Skewness value: 0.009 — nearly perfect symmetry around the mean
- Distribution is approximately normal — customers rate across the full scale without strong bias
- This is in contrast to typical 5-star rating systems where ratings cluster at extremes

Insight: The near-zero skewness (0.009) confirms the rating data is normally distributed, making it reliable for parametric statistical tests in future analysis.

4.4 COGS Distribution (Numerical)

The Cost of Goods Sold (cogs) column was analyzed via distplot and boxplot:

- Skewness: 0.893 — significantly right-skewed (positive skew)
- The majority of transactions involve lower-cost goods, with a long tail of high-value purchases
- The boxplot confirmed the presence of higher-end outlier transactions pulling the mean above the median

Insight: The right skew in cogs reflects the typical supermarket reality — most purchases are small/medium, with occasional large basket sizes significantly impacting averages.

5. Bivariate & Multivariate Analysis — Findings

5.1 COGS vs. Customer Rating (Num–Num)

A scatterplot of cogs (x) against Rating (y) revealed no discernible pattern or correlation between spending amount and customer satisfaction rating.

Insight: How much customers spend has no relationship with how they rate their experience. Service quality, product quality, and other factors drive ratings — not transaction value.

5.2 Gross Income vs. Customer Rating (Num–Num)

Similarly, a scatterplot of gross income vs. Rating showed no correlation, reinforcing the finding above.

Insight: Neither cogs nor gross income predicts rating. This is important for management — high-spending customers are not necessarily more satisfied, meaning every customer interaction matters equally.

5.3 Most Profitable Branch (Num–Cat + Multivariate)

A barplot of Branch vs. gross income revealed Branch C (Naypyitaw) as the most profitable, albeit marginally. Adding City as a hue variable confirmed the branch-city mapping:

Branch	City	Relative Gross Income	Rank
C	Naypyitaw	Highest	1st
A	Yangon	Similar to B	2nd
B	Mandalay	Similar to A	3rd

Insight: Branch C slightly outperforms on income despite having the fewest transactions (328) — suggesting higher average basket size or more premium product purchases in Naypyitaw.

5.4 Gender vs. Gross Income (Num–Cat + Multivariate)

A boxplot of Gender vs. gross income showed:

- Female customers spend slightly more on average than male customers

- The interquartile ranges overlap significantly — not a dramatic difference

Adding Customer type as hue revealed a more important insight:

- Member customers (loyalty card holders) spend more than Normal (non-member) customers
- This held true across both genders

Insight: Loyalty program membership is a stronger predictor of spending than gender. Expanding the member base is a higher-impact business lever than gender-targeted marketing.

5.5 Product Line Analysis

By Gross Income

Home and Lifestyle generates the highest gross income across all product lines. Electronic Accessories and Sports & Travel also perform well.

By Unit Price

Sports and Travel commands the highest average unit price, suggesting premium positioning. This means Sports & Travel's high revenue comes from price-per-unit rather than volume.

Product Line	Relative Gross Income	Relative Unit Price	Qty Sold
Home and lifestyle	Highest	Mid	911
Sports and travel	High	Highest	920
Electronic accessories	High	Mid-high	971 (most)
Food and beverages	Mid	Mid	952
Fashion accessories	Mid	Mid	902
Health and beauty	Lower	Mid	854 (least)

5.6 Payment Method by City — Cross-Tabulation (Cat–Cat Multivariate)

A cross-tabulation (pd.crosstab) of City vs. Payment produced a frequency matrix, visualized as a heatmap:

City	Cash	Credit Card	Ewallet
Mandalay	110	109	113
Naypyitaw	124	98	106
Yangon	110	104	126

Insight: Naypyitaw (Branch C) has the highest cash usage and lowest credit card adoption — suggesting a less digitally-engaged customer base. Yangon has the highest Ewallet usage, consistent with a more urban, tech-savvy population.

5.7 Quantity Sold by Product Line

A groupby sum of Quantity by Product line revealed:

- Electronic Accessories: 971 units — most purchased by volume
- Health and Beauty: 854 units — least purchased by volume
- All other lines fall in the 900–952 range — fairly balanced portfolio

Insight: Electronic accessories drive volume but may have lower margins per unit; Health & Beauty has the lowest volume but could have premium pricing opportunities.

6. Temporal Analysis — Feature Engineering

6.1 Sales by Day of Week

By mapping the Date column's day-of-week integer (0=Mon to 6=Sun) to day names, a new 'day_of_week' column was engineered and plotted:

Day	Sales Volume	Rank
Saturday	Highest	1st
Tuesday	Second highest	2nd
Monday/Wednesday/Friday	Moderate	Mid
Thursday/Sunday	Lower	Lower

Insight: Saturday dominates sales — expected for a retail environment. The unexpectedly high Tuesday sales (second place) is a notable finding, possibly driven by mid-week promotions or restocking cycles.

6.2 Sales by Month

Similarly, a 'month' column was derived from the Date and analyzed:

- January: Highest sales month across all three months
- February and March: Lower transaction volumes

Insight: January's high sales likely reflect post-holiday shopping patterns or New Year promotions. The declining trend through February–March warrants investigation into seasonal promotional strategies to sustain revenue.

Technical Note: Both datetime features were created using Pandas' dt accessor and .map() with custom dictionaries — a practical pattern for temporal feature engineering reusable on any date-indexed dataset.

7. Technical Skills Demonstrated

7.1 Data Profiling & Classification

- Programmatic categorical/numerical split using nunique() threshold loop
- Full schema inspection with .info(), .describe(), .dtypes, .shape
- isnull().sum() for null detection — confirmed zero missing values

7.2 Univariate Visualization

- sns.countplot() — frequency bar charts for categorical data
- sns.distplot() with KDE for distribution shape and skewness
- sns.boxplot() (single variable) for outlier detection in cogs
- .skew() — quantifying distribution asymmetry numerically (0.009 vs 0.893)
- .value_counts().plot(kind='bar') — quick categorical frequency charts

7.3 Bivariate & Multivariate Visualization

- sns.scatterplot(x, y) — correlation investigation between two numericals
- sns.barplot(x=cat, y=num) — mean comparison across categories
- sns.barplot(..., hue=third_variable) — 3-variable multivariate barplot
- sns.boxplot(x=cat, y=num, hue=cat2) — 3-variable grouped boxplot
- pd.crosstab(col1, col2) — frequency matrix for two categorical columns
- sns.heatmap(crosstab) — visual intensity map of cross-tabulation
- pl.xticks(rotation=60) — professional label rotation for readability

7.4 Feature Engineering & DateTime Handling

- parse_dates=['Date'] in pd.read_csv() — proper datetime ingestion
- .dt.dayofweek — extracting integer day of week from datetime
- .dt.month — extracting month integer from datetime
- .map(dict) — mapping integers to human-readable labels
- Creating new derived columns and validating with .head()

7.5 Aggregation

- data1.groupby('Product line').sum()['Quantity'] — grouped aggregation
- value_counts() for frequency distributions
- groupby with sort for ranked category analysis

8. Conclusion & Future Work

8.1 Summary of Findings

- All three branches have near-equal transaction volumes (~33% each)
- Ewallet is the most popular payment method; cash closely follows
- Customer ratings are normally distributed (skewness ≈ 0) — no bias
- COGS is right-skewed (0.89) — most purchases are small-to-medium
- Neither cogs nor gross income predicts customer rating scores
- Branch C (Naypyitaw) is the most profitable despite fewest transactions
- Member customers outspend non-members; females spend slightly more than males
- Home & Lifestyle generates most income; Sports & Travel has highest unit price
- Electronic Accessories sold in highest quantity (971 units)
- Naypyitaw prefers cash; Yangon leads in Ewallet adoption
- Saturday is peak sales day; January is peak sales month

8.2 Potential Enhancements

Enhancement	Description	Technique
Sales Forecasting	Predict monthly revenue for Apr–Jun 2019	ARIMA, Prophet, LSTM
Customer Segmentation	Cluster customers by spend & frequency	K-Means, RFM Analysis
Rating Predictor	Predict rating from product/branch/payment	Random Forest, XGBoost
Basket Analysis	Find frequently bought together products	Apriori, Association Rules
Interactive Dashboard	Live KPI monitoring for branch managers	Streamlit, Plotly Dash
Loyalty Program Impact	Deeper member vs. non-member revenue analysis	A/B Test Simulation

8.3 Learning Outcomes

This project significantly deepened my understanding of the relationship between analysis types and visualization choices. Learning to match the right chart (scatter for num-num, boxplot for num-cat, heatmap for cat-cat) to the right question is a core competency that this project reinforced through hands-on practice. The datetime feature engineering component also introduced a reusable pattern for temporal analysis applicable to any transactional dataset.

9. Project Information

Project Title	Supermarket Sales: Univariate, Bivariate & Multivariate EDA
Author	Prince Maheta
GitHub Repository	github.com/princemaheta2627-glitch/univariate_bivariate_multivariate_analysis_eda
Notebook File	9-univariate-bivariate-multivariate-answers.ipynb
Dataset	Supermarket Sales — Kaggle (1,000 rows, 17 columns, Jan–Mar 2019)
Branches	A-Yangon, B-Mandalay, C-Naypyitaw (Myanmar)
Platform	Kaggle Notebook / Jupyter (Python 3)
Languages & Tools	Python 3, Pandas, NumPy, Seaborn, Matplotlib
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